



02) Quadratic form and positive-definite matrix

Quadratic form is a second-degree polynomial function

Quadratic form function $V(x)$:

$$V(x) = x^T P x$$

x : an n -dimensional column vector

P : a symmetric matrix

$V(x)$: a scalar value.

Positive Definite Matrix:

P is positive definite if $V(x) > 0$ for any $x \neq 0$

Positive Semi-Definite Matrix:

P is positive semi-definite, if $V(x) \geq 0$ for any $x \neq 0$